PRAKTIKUM BASIS DATA

MODUL_08

Scripting_SQL_Create_

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Jurnal_08

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1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

JAWABAN

1.)

a.)

Query :

```
CREATE TABLE departments (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);
```

Hasil :

```
Table DEPARTMENTS created.
```

b.)

Query :

```sql
CREATE TABLE employees (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER(10,2),
    commission_pct NUMBER(5,2),
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES departments(department_id)
);
```

Hasil :

```
Table EMPLOYEES created.
```

c.)

Query :

```sql
CREATE TABLE suppliers (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    contact_email VARCHAR2(100)
);

CREATE TABLE products (
    product_id NUMBER PRIMARY KEY,
    product_name VARCHAR2(100),
    supplier_id NUMBER,
    price NUMBER(10,2),
    CONSTRAINT fk_supplier
        FOREIGN KEY (supplier_id)
        REFERENCES suppliers(supplier_id)
);

CREATE TABLE inventory (
    inventory_id NUMBER PRIMARY KEY,
    product_id NUMBER,
    quantity NUMBER,
    CONSTRAINT fk_product
        FOREIGN KEY (product_id)
        REFERENCES products(product_id)
);
```

Hasil :

```
Table SUPPLIERS created.

Table PRODUCTS created.

Table INVENTORY created.
```

2.)

Query :

```
BEGIN
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'DEPARTMENTS');
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEES');
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'SUPPLIERS');
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'PRODUCTS');
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'INVENTORY');
END;
/
```

Hasil :

```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.
```

3.)

Departments

Query :

```
INSERT INTO departments VALUES (207, 'HR', 101, 1000);
INSERT INTO departments VALUES (289, 'IT', 102, 1001);
INSERT INTO departments VALUES (315, 'Finance', 103, 1002);
INSERT INTO departments VALUES (222, 'Marketing', 104, 1003);
INSERT INTO departments VALUES (278, 'Sales', 105, 1004);
INSERT INTO departments VALUES (301, 'Logistics', 106, 1005);
INSERT INTO departments VALUES (256, 'Support', 107, 1006);
INSERT INTO departments VALUES (333, 'Engineering', 108, 1007);
INSERT INTO departments VALUES (244, 'QA', 109, 1008);
INSERT INTO departments VALUES (290, 'Admin', 110, 1009);
```

Hasil :

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Employees

Query :

```
INSERT INTO employees VALUES (601, 'Yensen', 'yensen@mail.com', 5000, 0.1, DATE '2026-12-01', 333);
INSERT INTO employees VALUES (602, 'Naia', 'naia@mail.com', 6000, 0.2, DATE '2026-12-02', 333);
INSERT INTO employees VALUES (603, 'Hary', 'hary@mail.com', 5500, 0.15, DATE '2026-12-03', 289);
INSERT INTO employees VALUES (604, 'Cia', 'cia@mail.com', 7000, 0.1, DATE '2026-12-04', 315);
INSERT INTO employees VALUES (605, 'Rehan', 'rehan@mail.com', 6500, 0.05, DATE '2026-12-05', 222);
INSERT INTO employees VALUES (606, 'Ilham', 'ilham@mail.com', 4800, 0.2, DATE '2026-12-06', 278);
INSERT INTO employees VALUES (607, 'Zulfa', 'zulfa@mail.com', 5200, 0.1, DATE '2026-12-07', 301);
INSERT INTO employees VALUES (608, 'Titu', 'titu@mail.com', 5100, 0.15, DATE '2026-12-08', 256);
INSERT INTO employees VALUES (609, 'Marta', 'marta@mail.com', 8000, 0.3, DATE '2026-12-09', 333);
INSERT INTO employees VALUES (610, 'Sari', 'sari@mail.com', 7500, 0.25, DATE '2026-12-10', 244);
```

Hasil :

```
1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

Commit complete.
```

4.)

Query :

```
BEGIN
    DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEES');
END;
/
```

Hasil :

```
PL/SQL procedure successfully completed.


PL/SQL procedure successfully completed.


PL/SQL procedure successfully completed.
```

5.)

Query :

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employees
WHERE department_id = 80;
```

Hasil :

```
View EMPVU80 created.
```

6.)

Query :

```
DESC empvu80;
```

Hasil :

```
View EMPVU80 created.

Name           Null?    Type
-------------- -------- -------------
ID_NUMBER      NOT NULL NUMBER
NAME           NOT NULL VARCHAR2(100)
SALARY                  NUMBER(10,2)
DEPARTMENT_ID           NUMBER
Name           Null?    Type
-------------- -------- -------------
ID_NUMBER      NOT NULL NUMBER
NAME           NOT NULL VARCHAR2(100)
SALARY                  NUMBER(10,2)
DEPARTMENT_ID           NUMBER
```

7.)

Query :

```
CREATE INDEX idx_dept_fk
ON employees(department_id);
```

Hasil :

```
Index IDX_DEPT_FK created.
```